

Amazon-KG: A Knowledge Graph Enhanced Cross-Domain Recommendation Dataset

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ABSTRACT

Cross-domain recommendation (CDR) aims to utilize the information from relevant domains to guide the recommendation task in the target domain, and shows great potential in alleviating the data sparsity and cold-start problems of recommender systems. Most existing methods utilize the interaction information (e.g., ratings and clicks) or consider auxiliary information (e.g., tags and comments) to analyze the users' cross-domain preferences, but such kinds of information ignore the intrinsic semantic relationship of different domains. In order to effectively explore the inter-domain correlations, encyclopedic knowledge graphs (KG) involving different domains are highly desired in cross-domain recommendation tasks because they contain general information covering various domains with structured data format. However, there are few datasets containing KG information for CDR tasks, so in order to enrich the available data resource, we build a KG-enhanced cross-domain recommendation dataset, named *Amazon-KG*, based on the widely used Amazon dataset for CDR and the well-known KG DBpedia. In this work, we analyze the potential of KG applying in cross-domain recommendations, and describe the construction process of our dataset in detail. Finally, we perform quantitative statistical analysis on the dataset. We believe that datasets like *Amazon-KG* contribute to the development of knowledge-aware cross-domain recommender systems. Our dataset has been released at <https://github.com/WangYuhan-0520/Amazon-KG-v2.0-dataset>.

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CCS CONCEPTS

• **Information systems** → **Recommender systems**; *Personalization*; *Task models*.

KEYWORDS

cross-domain recommendation, knowledge graph, knowledge-aware recommendation

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1 INTRODUCTION

In the past decade, recommender systems have been widely used in many businesses such as e-commerce, social networks, and media [28]. It collects user preferences automatically and filters vital information for them from the overloaded data. Cross-domain recommendation (CDR) is a promising way to alleviate data sparsity and cold-start issues by transferring knowledge (e.g., user interests or item attributes) from an auxiliary domain to a target domain, giving it more opportunities to explore users' broad interests and improving comprehensive recommendation performance [32].

The key issue in cross-domain recommendation is how to explore inter-domain correlations for effective information transfer. Existing methods can be divided into three categories: rating pattern-based transfer, embedding-based transfer, and context-based transfer[32]. Earlier work focused only on interaction data (mostly ratings), and these rating pattern-based methods[6, 25] tend to learn user ratings from the source domain and transfer the

obtained rating pattern to the target domain to improve recommendation performance. With the development of deep learning, embedding-based methods have made great progress. A popular scheme in embedding-based methods follows the idea of Embedding and Mapping (EMCDR) [8, 14, 16, 18, 33], which first utilizes collaborative filtering techniques to obtain user and item features under each domain separately, and then learns the inter-domain mapping functions to align the overlapping representations. But the limitation of this scheme is that its performance depends heavily on the domain overlapping information. Context-based methods utilize side information to support cross-domain recommendations. Such approaches [3, 9, 10, 27] first create links based on common context (e.g., social tags, item attributes, semantic attributes), and then they transfer knowledge of user/item across domains. However, the context information used is usually unstructured or the availability is limited to specific datasets/platforms. Moreover, such methods may ignore the intrinsic semantic relationships between different domains.

Knowledge Graph (KG) is a promising side information for cross-domain recommendation due to the following reasons. Firstly, KG is built based on massive amounts of real-world facts, and Encyclopedic KG integrates general knowledge from multiple domains that can support inter-domain correlation [5]. Secondly, the ontological nature and structured format of KG data can avoid extensive manual efforts [20]. To visualize the above points, we show an example by querying DBpedia¹ for information describing the item *Harry Potter* in the book domain, and Fig. 1 shows part of the queried information. We can see that there is a lot of semantic content related to the item: the subject includes “Magic” and “Curse”, and its theme involves “Fantasy literature”. In addition, KG describes the relevance through the edges between nodes in the graph. This structured format makes it possible to link items in other domains (e.g., the movie *Strike*) beyond the same domain (e.g., the book *The Wonderful Wizard of Oz*), thus discovering the cross-domain correlation. In addition to the advantages of the cross-domain nature and structured format of KG data, this example also demonstrates that KG can provide rich semantic information and reveal structural connectivity across domains.

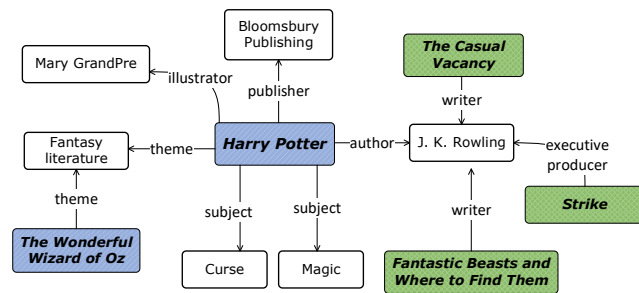


Figure 1: Partial information display of the example of correlated knowledge based on KG. Blue twill nodes represent book items and green mesh nodes represent movie items.

Therefore, we build a KG-enhanced cross-domain recommendation dataset named *Amazon-KG*, based on the widely used recommendation dataset Amazon² and the well-known KG DBpedia, aiming to introduce KG for cross-domain linking and knowledge enhancement. We select two popular domains in the cross-domain recommendation, *Books* and *Movies_and_TV* (denoted as *Movies* below for brevity). There are several difficulties in linking CDR items and KG entities, such as ambiguities in the names corresponding to multiple items belonging to different domains (duplicate names in other domains), or uncertainties in multiple entities that are queried based on a single item (a movie may have been remade multiple times). Without proper design, wrong linking may lead to false KG information as noise that affects the recommendation effect.

To address the above issues, we propose to build the dataset in a three-step manner. First, pre-link RS items to KG entities. We utilize the query language SPARQL³ to perform syntactic matching based on item titles and qualify the item types to avoid ambiguities of multiple items belonging to different domains. Second, with the linked entities from the previous step, we can query the item-related triples, which represent the auxiliary information about the item in KG. Third, since there are still similar item titles in the same domain, leading to the fact that an item may link to multiple entities, we further filter incorrect links based on item descriptions and queried KG information to achieve one-to-one matches between items and entities. Based on the constructed dataset, we perform several qualitative analysis experiments to discuss the effectiveness of cross-domain correlation via KG. With our knowledge graph enhanced CDR dataset, it is useful to develop an evaluation set for knowledge-aware CDR algorithms, and we believe that datasets like *Amazon-KG* facilitate the development of knowledge-aware cross-domain recommender systems.

2 EXISTING DATASETS AND METHODS

KG-based recommendation methods are well known for their potential to enhance accuracy and interpretability. Since KG contains rich semantic information, such methods can reason the intrinsic relationships between users/items by mapping items and their attributes to KG entities [5]. There have been many single-domain recommendation studies using KG as auxiliary information, focusing on improving the accuracy for preference prediction [21, 23] and work on improving the interpretability of results [11, 13]. Whereas in cross-domain recommendation, there are several KG-based methods [4, 17] that compute semantic similarity in an implicit manner (e.g., joint matrix factorization) to enhance preference analysis. It is worth noting that the current research on KG in CDR is not sufficient. Most of the existing work utilizes user-item interactions (e.g., ratings and clicks), while the few studies [22, 24] that have considered side information only analyze tags or reviews which are non-generic and unstructured. It suggests that the CDR community has the potential to make further progress in the exploration of auxiliary information. Therefore, we employ KG that contains general knowledge and has a structured data format, and build a KG-enhanced cross-domain recommendation dataset for facilitating knowledge-aware cross-domain recommendations.

¹<http://dbpedia.org>

²<https://nijianmo.github.io/amazon/index.html>

³<http://www.w3.org/TR/rdf-sparql-query>

We also notice some closely related studies [1, 4, 30] that are linking RS items to KB entities. However, we focus on cross-domain recommendation scenarios, which means that the dataset contains common users who have interactions in multiple domains and have explicit ID identifiers (e.g., user “A2S166WSCFIP5” in the Amazon dataset), instead of reordering the users from an ID of 1 that results in an inability to determine the user’s existence in other domains. Besides, we use the well-known KG DBpedia and the more widely used Amazon dataset in cross-domain recommendations to maximize the applicability of our dataset.

3 DATASET CONSTRUCTION

3.1 Preliminaries

3.1.1 RS dataset. Amazon dataset is a large-scale dataset of user-item interactions collected from the Amazon website and has been widely used in recommender systems. It covers a wide range of product categories such as music, movies, books, electronics, etc, and we have selected two popular domains in CDR research, *Books* and *Movies*. Amazon dataset provides user reviews and item metadata. User reviews describe the user’s preference for an item, containing information such as user ID, item ID, ratings, and time. Item meta-data characterizes the item in detail and contains information such as item ID, titles (names), and descriptions.

3.1.2 KG dataset. DBpedia is a multi-domain knowledge base that organizes structured knowledge from Wikipedia in a semantic-based manner. DBpedia has grown steadily with the maintenance of related associations and receives more attention than the similar Freebase [2], which is read-only and was shut down in 2016.

Knowledge graphs use the Resource Description Framework⁴ (RDF) to organize/model data, which typically describes facts consisting of three parts (also known as triples). Given E as the entities set and R as the relations set, we formally define the KG as $G_{kg} = \{(h, r, t) \mid h, t \in E, r \in R\}$, where the triple (h, r, t) represents an instance of “head_entity, relation, tail_entity” in KG, such as “*Harry Potter*, author, *J.K. Rowling*”. In addition, SPARQL is a query language and protocol developed for RDF, which we utilize in the following to link recommendation items to knowledge graph entities.

3.2 Collecting the Dataset

With the RS dataset and the KG dataset, we will perform data linking/matching between the two to explore RS-related KG information for knowledge enhancement. Next, we discuss the construction process of the *Amazon-KG* dataset through our proposed three-step manner.

3.2.1 Step 1. Pre-link Items to DBpedia Entities. Given a particular item, we identify the corresponding KG entity labeled with the item title by syntactic matching and type qualification. Specifically, we match the head entity (target item) by setting a SPARQL query with *rdfs:label*⁶ as relation and the item title as a tail entity. Due to ambiguities in the names of multiple items belonging to different domains, we qualify the type of retrieved items with *rdf:type*⁷ as

an additional relation. The following is an example query for the movie *Blue Thunder* :

```
PREFIX rdfs: <http://www.w3.org/2000/01/rdf-schema#>
PREFIX dbo: <http://dbpedia.org/ontology/>
PREFIX rdf: <http://www.w3.org/1999/02/22-rdf-syntax-ns#>

SELECT DISTINCT ?item WHERE {
  ?item rdf:type dbo:Film .
  ?item rdfs:label ?name .
  FILTER regex (?name, "Blue.*Thunder", "i") .
}
```

where *dbo:Film* is one of the DBpedia classes to represent the *Movies* types. We also list the item types and corresponding DBpedia classes in the dataset in the first and second columns of Table 1. It is worth noting that the example query returns two entities, *dbr:Blue_Thunder* and *dbr:Blue_Thunder_(2015_film)*⁸. To accurately match the item *Blue Thunder* to the KG entity, we will perform further filtering in Step 3 (Section 3.2.3).

3.2.2 Step 2. Retrieve the Related Triples. For each linked entity, we further query the item-related KG information, which are the triples of the linked entity as a head entity:

```
PREFIX dbr: <http://dbpedia.org/resource/>

SELECT DISTINCT ?p ?o WHERE {
  dbr:Blue_Thunder ?p ?o
}
```

Since our goal is item recommendation, we follow the previous work [4] to consider certain attributes under each domain for information filtering, thus focusing on KG information related to the common preferences of different users. We list the DBpedia classes related to the certain attributes selected for each domain of the dataset in the third column of Table 1. As an example, for the movie items, we focus on directors (*dbo:director*⁹), actors (*dbo:starring*), etc.

3.2.3 Step 3. Filter multiple entities for one item. After Step 1 and Step 2, we obtain the linked entities and related KG information for the items. However, we found that even after dealing with the ambiguities of item names in different domains, an item may still match multiple entities in the same domain. For example, remake movies have similar entity names, so the query for *Blue Thunder* returns two entities, *dbr:Blue_Thunder* and *dbr:Blue_Thunder_(2015_film)*. This is the case of inaccurate matches, and the wrong links will bring noise rather than auxiliary information, thus affecting the effectiveness of the recommendation.

To perform further accurate one-to-one matching of items and entities, we utilize the item descriptions in the RS dataset and the queried item-related information in the KG dataset. Specifically, for the *Movies* domain, since the item descriptions contain detailed information about the movie such as actors and directors, the query KG information also contains such information. Therefore, we select specific relations in the related triples of the target item and

⁴<https://www.w3.org/TR/rdf-primer>

⁶Namespace for rdfs, <http://www.w3.org/2000/01/rdf-schema>

⁷Namespace for rdf, <http://www.w3.org/1999/02/22-rdf-syntax-ns#>

⁸Namespace for dbr, <http://dbpedia.org/resource>

⁹Namespace for dbo, <http://dbpedia.org/ontology>

Table 1: The item’s domain and the corresponding DBpedia class. "Relation for itemType" qualifies the item type/domain to avoid name ambiguities for items in different domains; "Relation for item-related KG information" qualifies which type of item-related KG information is to be queried.

Domain	Relation for itemType	Relation for item-related KG information
<i>Movies</i>	dbo:Film, yago:Movie106613686 ⁵ , dbo:TelevisionShow, dbo:tvShow, dbo:televisionSeries	dct:subject, dbo:genre, dbo:author, dbo:creator, dbo:starring, dbo:presenter, dbo:director, dbo:executiveProducer, dbo:cinematography, dbo:openingTheme, dbo:basedOn, dbo:artist, dbo:notableWork, dbo:previousWork, dbo:subsequentWork, dbo:composer, dbo:producer, dbo:musicComposer, dbo:series, dbo:award, dbo:album
<i>Books</i>	dbo:Book, yago:Book102870092, yago:Book102870526, yago:LiteraryComposition106364329	dct:subject, dbo:genre, dbo:literaryGenre, dbo:author, dbo:creator, dbo:writer, dbo:series, dbo:portrayer, dbo:basedOn, dbo:notableWork, dbo:previousWork, dbo:subsequentWork, dbo:composer, dbo:musicComposer, dbo:artist, dbo:award, dbo:producer, dbo:album, dbo:director, dbo:starring

syntactically match the corresponding tail entities in the item description. These specific relations include the author (*dbo:author*) and publication time (*dct:subject*¹⁰) in the book domain, and director (*dbo:director*), actor (*dbo:starring*), and staging time (*dct:subject*) in the movie domain, where the format of the tail entities for publication time and staging time is such as *cat: 1999_British_novels* and *cat:2007_films*. With the above fine-grained verification, items and entities can eventually be mapped one-to-one with each other.

In addition to the above solutions, we also dealt with other issues of dataset collection. The anti-crawler mechanism of DBpedia restricts the query frequency and the maximum amount of data returned by a single query to 10,000¹¹. To deal with the query frequency problem, we set the hang time of the process through *time.sleep()* in Python code; for the maximum data limit, we utilize the keywords OFFSET and LIMIT in SPARQL to perform batch queries. We also solve problems that affect the syntactic matching results, such as case conversion, character escaping, and abbreviation reduction.

3.3 Final KG-enhanced Information

The final KG information we obtained contains the following two main types:

- Attributes: Related entities obtained from the considered type of item-attribute relations, e.g., the authors (*dbo:author*) and genres (*dbo:genre*, *dbo:LiterartGenre*) of a book.
- Related items: related entities derived from the type of item-item relations, e.g., a movie adapted from a novel (*dbo:basedOn*), a prequel/sequel to a movie (*dbo:previousWork*/*dbo:subsequentWork*).

4 STATISTICAL ANALYSIS

In this section, we analyze the statistics of the *Amazon-KG* dataset, focusing on the following questions: (1) What are the basic statistics of the knowledge graph-enhanced cross-domain recommendation dataset? (2) How to quantify the knowledge enhancement of KG for

different domains? (3) Beyond semanticity, how does KG correlate different domains in terms of structure?

Firstly, we summarize the basic statistics of the dataset in Table 2. We link 4896 movie items and 8696 book items from the RS dataset to the knowledge graph, and the queried KG information (triples) has 1062 common entities in both domains. We also make a quantitative analysis of the cross-domain connectivity in the datasets, where “*Freq. of common entity*” indicates the occurrence frequency of common entities in triples within a single domain, and “*CD-connectivity*” calculates the proportion of triples with common entities in the total triples. In addition to keeping the items in our dataset all KG-aligned, we also perform *k*-core filtering to ensure the quality of the RS dataset. This operation filters users and items with less than *k* interactions to alleviate data sparsity [19], and we set *k*=5 for the Amazon dataset. The Amazon-KG dataset we have processed is available at <https://github.com/WangYuhan-0520/Amazon-KG-v2.0-dataset> for further research.

Table 2: Statistics of the Amazon-KG dataset.

Domain	Movies	Books
Items	4,896	8,696
Users	5,648	10,485
Interactions	67,066	115,572
Sparsity	99.76%	99.87%
Entities	53,096	35,995
Relations	20	20
Triples	151,515	80,684
Overlap users	1,062	
Common entity	1,857	
Freq. of common entity	16.51%	14.97%
Cross-domain connectivity	15.97%	

Secondly, we analyze the enhancement of KG for recommendation datasets. We present the distribution of item-related triples of the two domains in Fig. 2 and 3, which indicate the number of items linked to that of triples. For a clear demonstration, we combine the number of triples into six bins. We can see that KG contains richer information related to the *Movies* domain, and the number

¹⁰Namespace for dct, <http://purl.org/dc/terms>

¹¹DBpedia has published some relevant information, detailed at <https://www.dbpedia.org/resources/sparql/>

of per-item enhanced knowledge/triples is majority distributed in the interval 10 ~ 40 with an average of 21.81. While the number of per-item augmented knowledge/triples in the *Books* domain is majority distributed in the interval 0 ~ 20 with an average of 8.00. This suggests that the *Movies* domain might benefit more from the KG. We analyze the possible reasons for this as Data Collection Bias & Community Engagement: Since DBpedia’s data source is Wikipedia, which serves as an open community editing platform, its content may exhibit inherent biases. Movies are generally easier to document and share, and movie enthusiasts and movie-related communities tend to be more active. As a result, there is a higher likelihood of having corresponding Wikipedia entries, thus enriching the available movie-related information. In the future, we will consider collecting more diverse and extensive datasets, developing algorithms that are robust to data imbalances, and other ways to deal with the presence of bias.

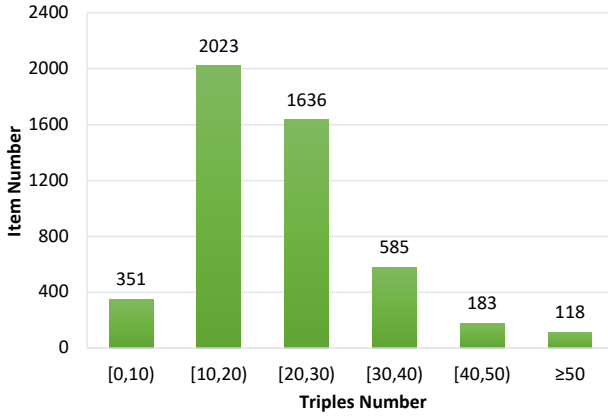


Figure 2: Distribution of related triples for items in *Movies* domain.

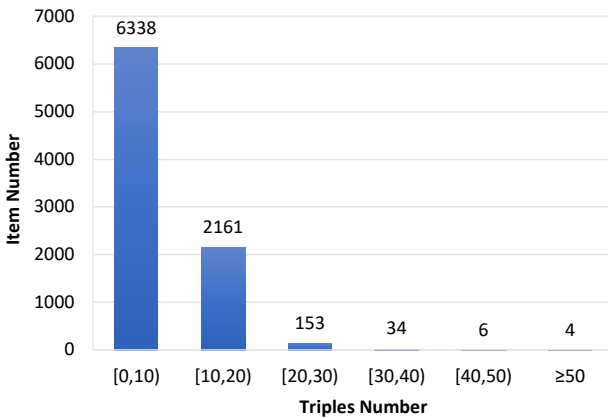


Figure 3: Distribution of related triples for items in *Books* domain.

Thirdly, we analyze several cases of common entities to visualize the cross-domain correlations enabled by the structural nature of

KG. It is not difficult to recognize that KG provides semantic information (e.g., a book’s genre and a movie’s theme) that can correlate different domains. Moreover, it is worth noting that the structural connectivity of KG enables cross-domain correlation as well. We choose five cases as shown in Fig. 4, containing five pairs of KG relations, namely *dbo:writer* & *dbo:author*, *dbo:executiveProducer* & *dbo:author*, *dbo:creator* & *dbo:author*, *dbo:director* & *dbo:author*, *dbo:presenter* & *dbo:author*, which link RS items from different domains. It can be seen that common entities such as *Arthur_Conan_Doyle* can link more than one item in a domain, making reasonable and meaningful connections between the items. In addition, it shows that KG is promising for solving the cold-start problem in RS. Assuming that a user has certain interactions in a domain, and when he is cold-starting in a new domain, cross-domain correlations based on KG can infer his preference tendencies, which helps the systems to provide recommendations for him in the new domain.

5 EXPERIMENTS

In this section, we conduct experiments on the KG-enhanced dataset to compare some existing recommendation algorithms.

5.1 Experimental Settings

5.1.1 Compared Methods. To demonstrate the practical implications of Amazon-KG, we compare the following methods, which fall into two categories: *single-domain* and *cross-domain* methods. *Single-domain* methods leverage data solely from the target domain for training, whereas *cross-domain* methods involve joint training using data from both the target and source domains, ultimately generating recommendations for the target domain.

Single-domain recommendation methods:

- **Pop**: a simple but common strategy to rank items by their popularity (measured by the number of users following).
- **BPR** [15]: Bayesian Personalized Ranking is a pairwise ranking method based on matrix factorization.
- **CKE** [26]: the first proposes to combine structural, textual, and visual information in a unified framework, where we only use KG information for fairness.
- **RippleNet** [21]: an end-to-end KG-based recommendation framework, which enriches user representations by adding items that are rooted in each user’s path.

Cross-domain recommendation methods:

- **EMCDR** [14]: learns latent factors for each domain separately based on matrix factorization and then uses a multi-layer perceptron to bridge the cross-domain latent factors.
- **SSCDR** [8]: an expanded EMCDR-based method, which trains a cross-domain mapping function based on the semi-supervised loss.
- **CoNet** [7]: first models interactions of two domains with two separate feed-forward networks, then transfers knowledge with a cross-connection network between the two base networks.
- **PPGN** [29]: uses stacked GCNs on the joint interaction graph to directly aggregate information from various domains to learn user/item features.



Figure 4: Case study of cross-domain structural connectivity based on knowledge graphs.

- **BiTGCF**[12]: utilizes GCN-based encoders to generate user/item features in both domains separately, and combines the user features via a feature transfer function to obtain common features.
- **CD-MFs**[4]: includes three KG-based cross-domain recommendation methods, which are based on matrix factorization model with different regularization designs, namely **SimMF**, **NeighborMF**, and **CentroidMF**. They regularize the model by computing the semantic similarity¹² of cross-domain item metadata in KG.

The implementation of CD-MFs is based on the original code¹³, and the other baselines are based on the *RecBole* library [31].

¹²We adopt the well-performing strategy *TF-IDF* from the paper for the computation of semantic similarity.

¹³<https://github.com/nachoft/cross-metadata-mf>

5.1.2 Experimental Evaluation. Following previous work, each domain data is split in the ratio of 6:2:2 for the training, validation, and test sets. To evaluate method performance, we employ four metrics that have been widely used in previous works [7, 21], including Recall, Hit Ratio (HR), Mean Reciprocal Rank (MRR), and Normalized Discounted Cumulative Gain (NDCG). The first two metrics are evaluated in terms of accuracy, focusing on whether the recommendation model is able to retrieve relevant items; the latter two focus not only on the relevance of the retrieved results, but also on the ranked position of these results in the recommendation list. We truncate the ranked list at 20 ($K = 20$), which means we only calculate metrics for the top 20 recommended items. We select the best evaluation results according to the highest Recall performance on the validation set.

Table 3: Experimental results of different methods. The best performance is highlighted in boldface and the second is highlighted by underlined. “†” after the name indicates that the method utilizes KG.

Dataset	Metric	Single-domain recommendation				Cross-domain recommendation							
		Pop	BPR	CKE†	RippleNet†	EMCDR	SSCDR	CoNet	PPGN	BiTGCF	SimMF†	NeighMF†	CentrMF†
Amazon Movie-Book	MRR	.0018	.0245	.0321	0.0316	.0247	.0285	.0187	.0335	.0427	.0963	<u>.0959</u>	.0954
	Recall	.0086	.0985	.1160	0.0893	.0967	.0974	.0542	.1497	.1467	<u>.1512</u>	.1514	.1514
	NDCG	.0029	.0370	.0458	0.0393	.0370	.0397	.0228	.0644	.0584	.0853	<u>.0851</u>	.0849
	Hit	.0150	.1421	.1659	0.1342	.1392	.1422	.0861	<u>.1800</u>	.2157	.0618	.0633	.0628
Amazon Book-Movie	MRR	.0134	.0300	.0372	.0324	.0331	.0209	.0300	.0220	.0403	.1051	<u>.1049</u>	.1051
	Recall	.0382	.1045	.1040	.0833	.1030	.0719	.0830	.1404	.1265	<u>.1622</u>	.1631	.1619
	NDCG	.0159	.0412	.0449	.0374	.0431	.0290	.0371	.0468	.0525	<u>.0916</u>	.0917	.0916
	Hit	.0660	<u>.1641</u>	.1640	.1328	.1638	.1094	.1284	.1404	.1987	.1091	.1097	.1084

5.1.3 Parameter Settings. For fair consideration, we fixed the latent dimension to 50 for all comparison baselines. The parameters of all these layers are initialized using Xavier normal implemented in PyTorch. The L2 regularizer coefficient is selected from $\{1e^{-3}, 1e^{-4}, 1e^{-5}\}$, the dropout rate is selected from $\{0.1, 0.2, 0.3, 0.5\}$, γ is selected from $\{0.1, 1, 10\}$, and the learning rate is selected from $\{0.001, 0.0005, 0.0001\}$. The training batch size is 40960, and the Adam optimizer is used to update all parameters. For all baselines, we perform fine-tuning on the datasets according to the hyperparameters provided in the original papers and present the final best results.

5.2 Results and Analysis

Table 3 shows the performance comparison results. We have the key observations in this experiment are as follows:

- Among the compared single-domain approaches, POP shows poor results because there is no consideration of the user’s personalized preferences. BPR can achieve better results. CKE shows promising performance, demonstrating the effectiveness of using KG as auxiliary information to improve recommendations. RippleNet also incorporates KG information, but this predefined path-based approach may have difficulty handling the noise generated when considering more entities.
- It is worth pointing out that cross-domain methods are not superior to single-domain methods in all scenarios, which differs from the results of previous work [4, 18, 22]. EMCDR, SSCDR and CoNet follow the idea of embedding and mapping, and by focusing only on domain overlapping information, they may be greatly affected by imbalanced data across domains, which results in negative transfer issues. PPGN and BiTGCF benefit from the superiority of GNN and have achieved good results in some cases.
- In contrast, all three KG-based methods of CD-MFs (SimMF, NeighborMF, CentroidMF) show competitive performance, which is attributed to the successful utilization of semantic information from KG. These three methods utilize KG to find relations between items in different domains and compute inter-domain item similarities, thus demonstrating the effectiveness of KG for exploring cross-domain correlations.

6 CONCLUSION AND FUTURE WORK

In this paper, we introduce a knowledge graph enhanced cross-domain recommendation dataset, *Amazon-KG*, which is based on the widely-used RS dataset Amazon and the well-known KG DBpedia. We detail the construction process of the dataset and extensively analyze its characteristics through statistics and case studies. We believe that such a dataset facilitates the development of knowledge-aware cross-domain recommender systems, and makes it easy to develop evaluation sets for knowledge-aware cross-domain recommender algorithms. In the future, we plan to extend our dataset to more domains to explore cross-domain scenarios, such as music, games, and so on.

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